Top Secret - Homework 5 UML Class Diagram

Relationship Key

- Uses
- Implements

TopSecret

+ main(args: String[]): void

UserInterface

- users: UserManagerInterface
- accessControl: AccessControlInterface
- store: SQLiteDataStore
- auditLog: AuditLogInterface

+ run(args: String[]): int
+ showFileList(): int
+ formatListEntry(number: int, fileName: String): String
+ parseFileNumber(text: String): int
+ fail(message: String): int

<<interface>>
AccessControlInterface

+ isAdmin(username: String): boolean
+ promoteUser(adminUsername: String, targetUsername: String): boolean
+ demoteUser(adminUsername: String, targetUsername: String): boolean
+ hasPermission(username: String, action: String): boolean

AccessControl

- admins: Set<String>
- users: Set<String>

+ AccessControl(admins: Set<String>, users: Set<String>)
+ isAdmin(username: String): boolean
+ promoteUser(adminUsername: String, targetUsername: String): boolean
+ demoteUser(adminUsername: String, targetUsername: String): boolean
+ hasPermisssion(username: String, action: String): boolean

Standard Users:
- Read items

Admins:
- Read items
- Create entries
- Create users
- Review audit log

Admins can promote standard users and demote other admins, but cannot demote themselves.

FileHandler

+ getFileLinesList(fileName: String, fileType: FileTypes): ArrayList<String>
+ listDataFiles(): ArrayList<String>

Cipher

- substitutions: HashMap<Character, Character>
+ Cipher(keyFileName: String)
+ decipher(cipherText: String): String

TopSecretException

+ TopSecretException(message: String)
+ TopSecretException(message: String, cause: Throwable)

<<interface>>
DataStoreInterface

+ listMissions(): List<Mission>
+ getMission(id: int): Mission
+ initDatabase(): void
+ importTsv(path: String): void

<<interface>>
AuthenticationInterface

+ createCredentials(user: String, pass: String): void
+ verifyLogin(user: String, pass: String): boolean
+ changePassword(newPass: String, confirmPass: String): void
+ credentialFileExists(): boolean

Authentication

+ createCredentials(user: String, pass: String): void
+ verifyLogin(user: String, pass: String): boolean
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+ credentialFileExists(): boolean

<<interface>>
UserManagerInterface

+ getUsers(): Set<String>
+ getUsername(): String
+ addUser(username: String): void
+ userExists(username: String): boolean

UserManager

- store: SQLiteDataStore

+ UserManager(store: SQLiteDataStore)
+ getUsers(): Set<String>
+ getUsername(): String
+ addUser(username: String): void
+ userExists(username: String): boolean

First account created is an admin.
All future accounts are standard users by default.

<<interface>>
AuditLogInterface

+ logAction(username: String, action: String): void
+ getLogs(): List<String>

AuditLog

- logFile: String

+ AuditLog(logFile: String)
+ logAction(username: String, action: String): void
+ getLogs(): List<String>

SQLiteDataStore

+ listMissions(): List<Mission>
+ getMission(id: int): Mission
+ initDatabase(): void
+ importTsv(path: String): void

Mission

- id: int
- title: String
- brief: String
- date: String